Practical 2: Particle Settling through a Fluid

(The 'Intelligent' Use of Equations to Quantify Sedimentary Processes)

… if you measure that of which you speak, you know something of your subject; but if you cannot measure it, your knowledge is meagre and unsatisfactory.

Lord Kelvin

BACKGROUND AND RELEVANCE

Particle settling is extremely important in sediment transport and deposition. As soon as grains are entrained into a flow, they begin to fall back to the bed. The distance that they travel down current before they again come to rest on the bed depends on the drag and lift forces exerted by the current (including turbulence) and the *settling velocity* of the particles.

Sediment particles also settle through static or very slowly moving fluids, for example silt from river mouth plumes entering the ocean, dead pelagic shells sinking through the ocean, or ash and pumice from a volcanic eruption.

As we discussed last week, a knowledge of particle settling provides the basis for a number of hydraulic methods for the measurement of particle size.

However, the concepts you will think about in this class have a broad application that extends beyond sedimentology. For example, settling rates are often used in the food industry for separating dirt and debris from incoming raw materials or in the environmental sector for studying the dispersal of airborne pollutants such as smoke particles.

AIMS

In this practical class we will use experimentation to

- to gain a fundamental understanding of the key physical properties of fluids and sediment particles (and a knowledge of their units)
- to gain an appreciation of the variables that influence the rate at which a sediment particle will settle through a still fluid (in particular those which resist and those which drive particle settling)
- to develop skills in experimental design (i.e. to develop a hypothesis and to plan and execute experiments to test those hypotheses)

At the end of this set of experiments you should be able to:

- name what physical attributes of a fluid and a particle are important in controlling the rate at which a particle settles
- explain what aspect of flow around a grain the grain Reynolds number describes
- describe the range of grain sizes over which Stokes' Law is applicable, elucidate why it breaks down, and be able to calculate settling velocities using it
- predict the significance of different rates of particle settling on the textural attributes of sedimentary deposits (e.g. grain size sorting within beds)

Terms you should be conversant with at the conclusion of this class: mass, weight, acceleration due to gravity, density, specific gravity, buoyancy, dynamic (molecular) viscosity, kinematic viscosity, settling velocity, terminal velocity, flow separation, drag, grain Reynolds number.

Part A: An experimental exploration of the controls on particle settling

APPARATUS

Two one-litre measuring cylinder each containing either a Glycerol or glycerol/water mixture. Stop watch and thermometer. Tape measure. Glass spheres of different sizes. Caliper gauge.

PROCEDURE AND OBSERVATIONS

You will complete the experiments in research teams of 6 people. Each team will split into two groups of three. One group will complete the procedure outlined below using Cyclinder 1 and the other group will use Cylinder 2. Cylinder 1 is filled with glycerol. Cylinder 2 is filled with a 50:50 mixture of water and glycerol. After collecting and analsysing the data, two groups will then compare their results. Read through the experiment procedure outlined below and decide what role each team member will play before you begin. Use your team members effectively in order to complete this exercise efficiently.

Select 4-5 spheres of different diameters and place each in a separate weighing boat. Taking care not to lose the smaller ones. Note the average diameter of each grain fromt eh information provided, and put the boats in size order.

Locate the 800mm line on the cyclinder (about one-third of the way down the cyclinder). Measure the height of the column of fluid down to the level of this mark, and also measure the temperature of the fluid.

Progressively release each grain one at a time into the fluid and time their decent between the 800mm line and the bottom of the cylinder (five in total). Tabulate and plot the results as you go- do not leave this to the end!

Observe the falling sphere: Does each grain appear to fall steadily?

Tabulate the time for descent against the measured average particle diameter. Calculate from the descent time and the height of the gylcerol column the particle terminal settling veolocity. Plot on the log-log graph paper the settling velocity against the particle diameter. **Both groups should plot their results on the same graph.**

RESULTS

Fluid in cylinder:

Viscosity of fluid (kg/m.s):

Density of fluid (kg/m^3):

Temperature (deg. C):

Particle diameter (mm)	Distance (m)	Time (s)	Settling velocity (m/s)

Discuss your results within your research team of 6 to answer the following questions.

What are the form and slope of the two graphs?

What is the general velocity-diameter relationship?

Why do the grains sink, and why is their fall steady?

Which fluid yields the higher plot?

What differences do you note in the physical characteristics of the pure glycerol and the 50:50 mixture?

How do you suppose these differences bear on the relative position of the plots?

CLASS ANALYSIS

The grains exist in the gravity field and sink because in both cases they are more dense than the fluid. The weight-force driving the motion of a grain is equal to the particle weight less, by the principle of Archimedes, the weight of the fluid displaced. That is

$$\text{Weight force} = \frac{4}{3}\pi\frac{D^3}{8}g(\rho_s - \rho) \qquad \text{Equation 1}$$

where:

g	acceleration due to gravity, L t^{-2}
D	a linear dimension, grain diameter, L
ρ_s	density of solid, sediment grain, M L^{-3}
ρ	density of fluid, water, M L^{-3}

You saw that the particles tended to settle steadily, after a short initial period of acceleration. At this point the particle maintained a constant velocity (i.e. it no longer accelerates) called the terminal settling velocity. Newton's First Law states that "Every body persists in its state of rest or of uniform motion in a straight line unless it is compelled by forces impressed upon it". Therefore, this requires that each grain is also operated on by a force equal and opposite to its downward-acting weight. What is this force? The only possibility is the resisting force, or drag, afforded by the fluid.

That is
Fluid drag = Particle projection area x Drag coefficient x Drag force per unit area

$$F_R = \frac{C_D\pi D^2\rho\omega^2}{8} \qquad \text{Equation 2}$$

where C_D is an unknown drag coefficient and ω the steady terminal settling velocity.
You can see from Eqn 2 that F_R will increase with the square of the velocity of a settling particle.

In the case of negligibly small inertial forces, Stokes calculated theoretically that $C_D = 24/Re$, where the Reynolds number is

$$Re = \frac{\rho D\omega}{\mu}$$

Using Stoke's result for C_D in Equation 2, the terminal settling velocity occurs when

$$\underbrace{3\mu\pi D\omega}_{F_R} = \underbrace{\frac{4}{3}\pi\frac{D^3}{8}g(\rho_s - \rho)}_{F_G}$$

Rearranging the equation in terms of settling velocity the expression becomes:

$$\omega = \frac{D^2 g(\rho_s - \rho)}{18\mu}$$

where:

ω	settling velocity, L t^{-1}
g	acceleration due to gravity, L t^{-2}
D	a linear dimension, grain diameter or depth of flow, L
ρ_s	density of solid, sediment grain, M L^{-3}
ρ	density of fluid, water, M L^{-3}
μ	dynamic (molecular) viscosity M L^{-1} t^{-1}

This equation is known as **Stokes' Law** and is used for calculating the settling velocity of particles.

Part B: Application of Stokes' Law to the calculation of particle settling velocity in water

APPARATUS

1.5 m tall Perspex tube filled with water (Tube 1). Stop watch and thermometer. Tape measure. Particles labelled Samples A, B and C. Calculator.

PROCEDURE AND OBSERVATIONS

1. **Using Stokes' Law**, calculate the settling velocity of the minimum and maximum sized particles within Sample A.

Assume that the average water temperature is 20°C, with a viscosity of 1.003x10^{-3} kg s^{-1} m^{-1} and a density of 998 kg m^{-3}. The grain density can be assumed to be 2650 kg m^{-3} (i.e. quartz). Gravity is 9.8 m s^{-2}. The sample contains a range of grain sizes varying between 0.1 mm and 0.3 mm diameter. Calculate the settling velocity for both the minimum and maximum grain sizes.

$\mathbf{D_{min}} = 0.1$ mm $\mathbf{D_{max}} = 0.3$ mm

(remember to convert your grain diameter measurement into m before using it to calculate the settling velocity. The settling velocity should be in terms of ms^{-1}).

$\omega =$ $\omega =$

Next, **experimentally measure** the average settling velocity of Sample A in Tube 1 (water). Remember to show your velocity in terms of ms^{-1}.

$\omega =$

How does this value compare to the velocities you calculated using Stokes' Law? Hint: *Does it fall between the minimum and maximum values?*

Answer:

2. **Using Stokes' Law**, calculate the settling velocity of Samples B and C in Tube 1.

For Tube 1: assume that the average temperature is 20°C, a viscosity of 1.003x10^{-3} kg s^{-1} m^{-1} and a density of 998 kg m^{-3}.

Estimate the grain density of Samples B and C from their material properties. Be sure to convert your density to kg m^{-3} before using it to calculate the settling velocity with Stokes' Law.

Acceleration due to gravity is 9.81 m s^{-2}.

Sample B $D = 3.7$ mm ***Sample C*** $D = 16.0$ mm

$\omega =$ $\omega =$

Stop and think about these values……

For comparison, calculate the velocity of one of the following:

1. The Manchester-Euston Virgin train?
2. A shot from a cannon?
3. The speed that your demonstrators Drew or Sam can run down the corridor?

Answer:

_____m s^{-1}

What I am asking you to think about, is, does the value you calculated for the settling velocity of these particles using Stokes' Law seem reasonable to you?

Circle Yes or No. Why? What bothers you about this value?

Next, **experimentally measure** the settling velocity of Samples B and C in Fluid 1.

Sample B $\omega =$ ***Sample C*** $\omega =$

How does this measured value compare to the velocity you calculated using Stokes' Law?
(Hint: Is the measured velocity greater or less than the value calculated using Stokes' Law?)

Answer:

Now ponder: why does Stokes' Law appear to correctly calculate the settling velocity of the very fine sand but not the coarser grain sizes? Why do coarser grain sizes settle more slowly than predicted by Stokes' Law?

ANALYSIS

The unrealistic result is the consequence of the inherent assumptions in Stokes' Law about what contributes to the drag or fluid resistance force (F_R). This exercise illustrates the clear need to understand a formula *before* you use it to make calculations or else your results will be nonsense!

Buoyancy is one contributing factor to the fluid resistance force on a particle (the upward directed force which slows the particle descent), as we saw in an extreme case when you tried to get the green balls to sink! However, a particle moving down through a fluid also generates resistance due to the shape and size of the particle, and the fluid's viscosity and density. We saw in Part A that to take care of this there is a term within the derivation of the F_R called the drag coefficient, C_D.

In Stokes' Law the C_D is based on the dynamics of the flow around the particle i.e. whether the flow around the grain is turbulent or laminar. This doesn't mean the whole flow has to be laminar- the grain is only aware of the flow directly around it. Even in cases where water is turbulent (which is usually the case) if a grain is small enough in diameter relative to the characteristic diameter of the turbulence, then the grain will act as if it is locally in laminar flow (or if you prefer, the grain thinks it is in laminar flow).

So how can we determine whether the flow around the grain is laminar or turbulent?
In the lecture we saw that we can determine if a flow is laminar or turbulent using the Reynold's number. In this application of the Reynolds number the length scale, D, is the grain diameter rather than water depth. Consequently this is called the grain Reynolds number (Re_{gr}) and is a measure of the degree of turbulence at the grain-fluid boundary (i.e. if the boundary layer around the particle is turbulent or laminar).

In equation form we can say the same thing as:
Grain Reynolds Number is equal to the settling velocity times the grain diameter divided by the kinematic viscosity of the fluid (i.e. molecular viscosity divided by the density), or

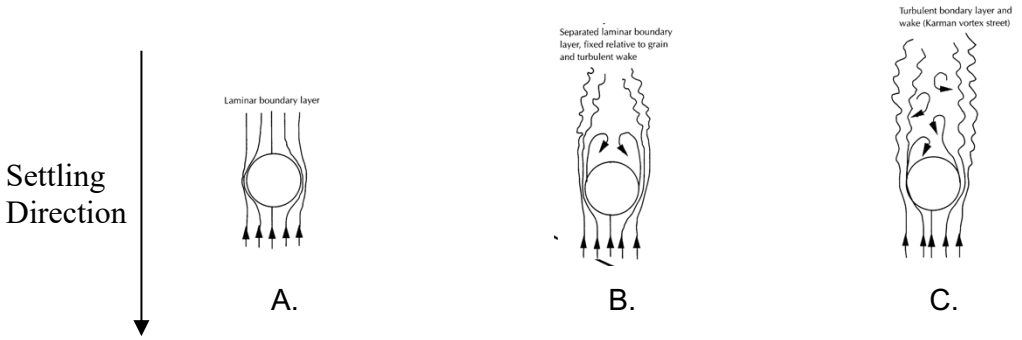
$$\mathrm{Re}_{gr} = \frac{\omega D}{u} = \frac{\omega D \rho}{\mu}$$

A. For Re_{gr} <1 there is a laminar boundary layer in the vicinity of the grain and no turbulent eddies.

Therefore, the flow around the grain can be viewed as having laminar behaviour.

B. For Re_{gr} >1<1000 the laminar boundary around the grain separates and there is a turbulent wake.

C. For Re_{gr} >1000 there is a turbulent boundary layer in the vicinity of the grain and a turbulent wake.



Calculate Re_{gr} for Samples A, B and C in Tube 1 to determine the nature of the flow around the grain

Note: Use your experimentally measured settling velocities to calculate the Re_{gr} here.

Sample A Re_{gr}=

The flow around the grain is _____

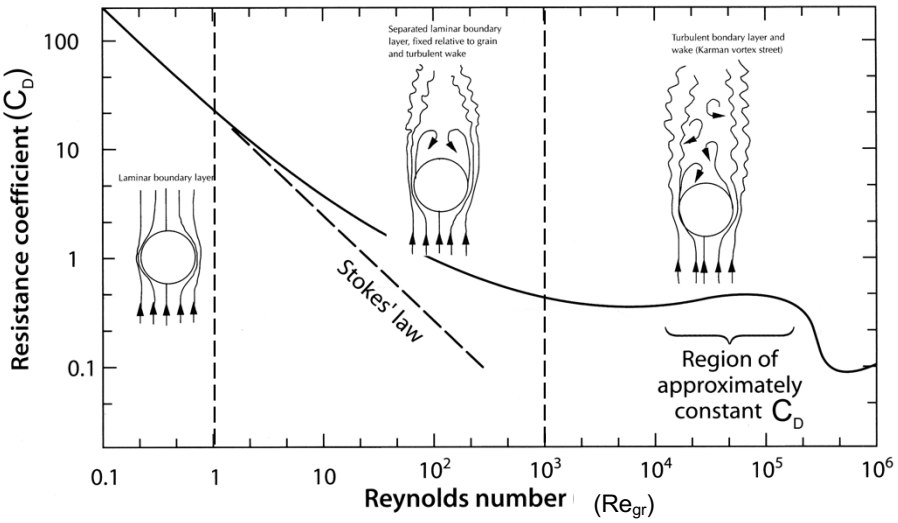
Sample B Re_{gr}=

The flow around the grain is _____

Sample C Re_{gr}=

The flow around the grain is _____

Locate your Re_{gr} values on the curve below with a vertical line



(after Hsu (2004) and Leeder (1999))

Examine the graph above. The graph shows the relationship between the resistance coefficient (C_D) and Reynolds number, calculated on the basis of experimental data.

In Stokes' Law it is assumed that the C_D is inversely proportional to Re_{gr} (i.e., $C_D = 24/Re$). However, the graph shows us that this is only true at low values of Re_{gr} when flow around the grain is laminar. This low Reynolds number range is called the Stokes range. So, if the grain is small and/or settles slowly and/or the fluid behaves viscously then the grain settles as if it were moving down through a laminar flow, and the velocity calculated from Stokes' Law agrees well with the measured settling velocity.

To verify this compare the settling velocity for Sample A that you experimentally measured with the velocity you calculated using Stokes' Law.

The graph also shows us that this linear relation does not hold true for turbulent flows with large values of Reynolds numbers. In these flows resistance is not dominated by viscous resistance (or friction) generated at the grain boundary but by turbulence throughout the flow (i.e. inertial resistance is dominant over viscous resistance). The graph shows that at high and very high values of Re the value of C_D is constant (approx. 0.4 for spheres), and that the value of C_D is higher than would be predicted by Stokes' Law, resulting in greater total. Thus, the particle settles more slowly than would be predicted by Stokes' Law.

Stokes' Law, therefore, only applies to sediment grains with diameters less than approximately 0.1 mm in water and 0.06 mm in air.

For different regions of settling behaviour, we need to use different settling laws.

At low Reynolds numbers (ie for small ω and grain diameters (D), the settling is viscous and we can use Stokes' Law to determine ω. Basically we could use Stokes' Law for quartz grains settling in water on Earth up to about 100μm diameter.

Examples:

D= 1 μm =10^{-6} m $\Rightarrow$ $\omega \cong 10^{-6}$ m/s (i.e. falls ~ 1 diameter in 1 second)
(clay size)

D= 10 μm =10^{-5} m $\Rightarrow$ $\omega \cong 10^{-4}$ m/s (i.e. falls ~ 10 diameters in 1 second)

D= 100 μm =10^{-4} m $\Rightarrow$ $\omega \cong 10^{-2}$ m/s = 1 cm/s (i.e. falls ~ 100 diameters in 1 second)

At intermediate Reynolds numbers, recourse must be had to empirical formulae:
For 100μm<D<1 mm (medium to coarse sand) the settling velocity depends on both viscosity and pressure, but over this range there is an approximately linear relationship between ω and D:

D= 0.25 mm $\omega \cong 3$ cm/s
D= 0. 5 mm $\omega \cong 7$ cm/s
D= 1 mm $\omega \cong 13$ cm/s

} range for med-coarse sand: 5-15 cm/s (roughly linear)

At sufficieintly high Reynolds numbers ($Re > 10^3$) we use another law in which the settling velocity varies with $D^{1/2}$:
For D>1 mm settling is resisted by unequal 'impact' pressure across the grain. You can use:

pressure force = immersed weight

$$\frac{1}{2}\rho C_D \omega^2 (\frac{\pi}{4} D^2) = g\rho(s-1)(\frac{\pi}{6} D^3)$$

$$\omega = \left[\frac{4}{3}\frac{(s-1)}{C_D} gD\right]^{\frac{1}{2}}$$

$$\omega \approx 5\sqrt{D}$$

e.g. D= 0.01 m $\Rightarrow$ $\omega \cong 0.5$ m/s
but you will rarely have cause to use this because grains this large are rarely suspended.

Equations to use to calculate the settling velocity of different sized particles in water.

What size particles could you apply Stokes' Law to if you wanted to calculate the settling velocity of feldspar crystals in a magma chamber? Larger or smaller grain diameters than in water?

Part C: Influence of settling velocities on sediment sorting and textures

Many textural attributes of sedimentary deposits, such as grain size grading, are probably a function of settling velocity.

PROCEDURE AND OBSERVATIONS

Blue-caped Cylinder 1

With the lid of the cylinder on tightly, shake the cylinder vigorously (i.e. create turbulence so that all the grains are entrained into the flow). Let the cylinder sit. Wait a few minutes until the grains have settled. The fluid in this cylinder is water

This distribution is known as **normal grading**. If a flow suddenly stops moving, all the grains will fall out and normal grading will occur. Because the mass of a particle (which drives the downward motion of the grain) increases with the cube of the grain diameter, but the resisting forces only increase with the square of the grain diameter (or its surface area), larger particles settle more quickly than small particles. Very small particles can be kept in suspension indefinitely by static charges and Brownian motion.

Blue-capped Cylinder 2

With the lid of the cylinder on tightly, shake the cylinder vigorously (i.e. create turbulence so that all the grains are entrained into the flow). Let the cylinder sit. Wait a few minutes until the grains have settled. The fluid in this cylinder is a glyercol-water mixture. Compare the distribution of the large and small particles in this cylinder to Cylinder 1. The deposits in Cylinder 2 are matrix-supported.

ANALYSIS

$$\omega = \frac{D^2 g(\rho_s - \rho)}{18\mu}$$

Look at the equation and realise:

1. a slight increase in grain size leads to normal grading
2. an increase in viscosity and/or an increase in ρ can lead to a matrix-supported deposit (i.e. large clasts appear to be floating in a matrix of fine grained mud) as opposed to grain-supported deposit where each large clast is resting on another large clast.

Letter	Upper case	Lower-case	Letter	Upper-case	Lower-case
alpha	A	α	nu	N	ν
beta	B	β	xi	Ξ	ξ
gamma	Γ	γ	omicron	O	ο
delta	Δ	δ	pi	Π	π
epsilon	E	ε	rho	P	ρ
zeta	Z	ζ	sigma	Σ	σ
eta	H	η	tau	T	τ
theta	Θ	θ	upsilon	Y	υ
iota	I	ι	phi	Φ	φ
kappa	K	κ	chi	X	χ
lambda	Λ	λ	psi	Ψ	ψ
mu	M	μ	omega	Ω	ω

Useful formulae for calculating areas and volumes of objects

Cube

Surface area: $A = 6Wh$

Volume: $V = L * W * h$

Spheres

Surface area: $A = 4\pi r^2$

Volume: $V = \frac{4}{3}\pi r^3$

Cylinder

Surface area: $A = 2\pi r^2 + 2\pi rh$

Volume: $V = \pi r^2 h$